Exercise 1

$\varinjlim$ is a functor.

Exercise 2

(1) Let $x \in X$ and $[(U, s)] \in \ker \varphi_x$. Then $[(U, \varphi_U(s))] = 0$ in $\mathcal{G}_x$, i.e. there exists an open neighbourhood $x \in V \subseteq U$ s.t. $0 = \varphi_U(s)|_V = \varphi_V(s|_V)$. But φ_V is injective by assumption, so $s|_V = 0$. Hence $[(U, s)] = [(V, s|_V)] = 0$ and φ_x is injective. Since x was arbitrary, we conclude that φ is injective.

For a counterexample of the reverse implication, let $|X| = 2$ with the discrete topology, let $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$. Since all restrictions are trivial, this is clearly a presheaf. Furthermore, for any $x \in X$, $\{x\}$ is final in the directed system $U \ni x$, so $\mathcal{F}_x = \mathcal{F}(\{x\}) = 0$. Now let $\mathcal{G} = 0$ on X and consider the unique morphism $\mathcal{F} \to \mathcal{G}$ (again it is obvious that this is a morphism since all restrictions are trivial). Since all stalks are trivial, this is injective, but $\mathbb{Z} = \mathcal{F}(X) \to \mathcal{G}(X) = 0$ is not.

(2) This was proven as part of theorem 1.11.

(3) Let $x \in X$ and $[(U, t)] \in \mathcal{G}_x$. Since φ_U is surjective, we can find a preimage $s \in \mathcal{F}(U)$ of t. Then $\varphi_x([(U, s)]) = [(U, \varphi_U(s))] = [(U, t)]$, so φ_x is surjective. Since x was arbitrary, by definition φ is surjective.

For a counterexample of the converse, consider the presheaves of (1) and the unique morphism $\mathcal{G} \to \mathcal{F}$. For a counterexample involving sheaves, consider $\underline{\mathbb{Z}} \to i_{x,*}\mathbb{Z} \oplus i_{y,*}\mathbb{Z}$, $x \neq y \in X$, X a T_0-space, where $i_{x,*}A$ denotes the skyscraper sheaf, the sheafification of the presheaf $U \mapsto A$ if $x \in U$ and $U \mapsto 0$ otherwise.

Exercise 3

(1) The restriction maps of $\ker \varphi, \operatorname{im} \varphi, \operatorname{coker} \varphi$ are restrictions/quotients of the restriction maps of $\mathcal{F}$ or $\mathcal{G}$. Hence they satisfy the same identities as the original maps.

(2) Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ an open cover. Let $s \in \ker \varphi(U)$ s.t. $s|_{U_i} = 0$ for all $i \in I$. Since $\ker \varphi \subseteq \mathcal{F}$ and $\mathcal{F}$ is a sheaf, this immediately implies $s = 0$ in $\mathcal{F}(U)$.

Now let $s_i \in \ker \varphi(U_i)$ be given, satisfying $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$. Again since $\mathcal{F}$ is a sheaf, these glue to a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. It remains to check $s \in \ker \varphi(U)$. One computes

$$\varphi_U(s)|_{U_i} = \varphi_{U_i}(s|_{U_i}) = \varphi_{U_i}(s_i) = 0,$$

and using that $\mathcal{G}$ is a sheaf, we conclude $\varphi_U(s) = 0$.

Exercise 1

Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover of U. Let $s = s_1 \oplus s_2 \in (\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$ be such that $s|_{U_i} = 0$ for all $i \in I$. By definition of the restriction map, $0 = s|_{U_i} = s_1|_{U_i} \oplus s_2|_{U_i}$ implies $s_j|_{U_i} = 0$ for $j \in \{1, 2\}$. Since this holds for all $i \in I$, and $\mathcal{F}$ and $\mathcal{G}$ are sheaves, we conclude $s_j = 0$, and thus $s = s_1 \oplus s_2 = 0$.

Now let $s_i \oplus t_i \in \mathcal{F}(U_i)$ be given, s.t. $(s_i \oplus t_i)|_{U_i \cap U_j} = (s_j \oplus t_j)|_{U_i \cap U_j}$ for all $i, j \in I$. Again, by definition of the restriction maps, this means $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ and $t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}$, hence these glue to elements $s \in \mathcal{F}(U)$, $t \in \mathcal{G}(U)$ satisfying $s|_{U_i} = s_i$ and $t|_{U_i} = t_i$. Then $(s \oplus t)|_{U_i} = s|_{U_i} \oplus t|_{U_i} = s_i \oplus t_i$, so $s \oplus t \in (\mathcal{F} \oplus \mathcal{G})(U)$ is the desired element.

Exercise 2

(1) Let A be an abelian group, X a topological space, $x \in X$. Write $i : \{x\} \hookrightarrow X$ for the inclusion. Then

$$i_* \underline{A}(U) = \underline{A}(i^{-1}(U)) = \begin{cases} 0 & \text{if } i^{-1}(U) = \emptyset \Leftrightarrow x \notin U, \\ A & \text{otherwise} \end{cases} = {}^x\mathcal{F}(U),$$

so ${}^x\mathcal{F} = i_* \underline{A}$ (the restriction maps clearly agree), and the result follows from exercise 3.

(2) Let X be a connected T_1-space, $x \neq y \in X$ and $0 \neq A$ an abelian group. We are looking at the natural map $\varphi : \underline{A} \to {}^x\mathcal{F} \oplus {}^y\mathcal{F}$. Clearly $\varphi_X : A \to A \oplus A$, $a \mapsto (a, a)$ is not surjective.

First let $z \notin \{x, y\}$. Then the set of open neighbourhoods U of z not containing x, y is cofinal in the set of all such neighbourhoods, so

$$({}^x\mathcal{F} \oplus {}^y\mathcal{F})_z = \varinjlim_{U \ni z} ({}^x\mathcal{F} \oplus {}^y\mathcal{F})(U) \cong \varinjlim_{U \ni z; x, y \notin U} ({}^x\mathcal{F} \oplus {}^y\mathcal{F})(U) = \varinjlim_{U \ni z; x, y \notin U} 0 = 0,$$

and the map $\underline{A}_z \to 0$ must be surjective.

Now let $z \in \{x, y\}$, wlog $z = x$. Similarly to above, we can compute φ_x as the limit over the cofinal set of open neighbourhoods U of x not containing y. But on any such U, $\varphi_U : A \to A \oplus 0$ is the identity by definition, hence also $\varphi_x = \mathrm{id}_A : A \to A$.

Exercise 3

$f_* \mathcal{F}$ is clearly a presheaf with the restrictions from $\mathcal{F}$. Let $U \subseteq Y$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover. Let $s \in f_* \mathcal{F}(U)$ satisfy $s|_{U_i} = 0$ for all $i \in I$. Note that $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is still an open cover since f is continuous. So $s \in \mathcal{F}(f^{-1}(U))$ is an element which is 0 when restricted to any open in that cover, hence $s = 0$ since $\mathcal{F}$ is a sheaf.

Now let $s_i \in f_* \mathcal{F}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ be given. Again, since $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is an open cover, these glue to an element of $\mathcal{F}(f^{-1}(U)) = f_* \mathcal{F}(U)$ satisfying $s|_{U_i} = s_i$, as desired.

Exercise 4

(1) Since $IJ \subseteq I \cap J$, by the lecture we have $V(I \cap J) \subseteq V(IJ) = V(I) \cup V(J)$. Conversely, since $I \cap J \subseteq I$, one gets $V(I) \subseteq V(I \cap J)$ and symmetrically $V(J) \subseteq V(I \cap J)$, hence also $V(I) \cup V(J) \subseteq V(I \cap J)$.

(2) We have

$$\begin{aligned}\mathfrak{p} \in V\left(\sum_\alpha I_\alpha\right) &\iff \sum_\alpha I_\alpha \subseteq \mathfrak{p} \iff I_\alpha \subseteq \mathfrak{p},\ \forall\alpha \\ &\iff \mathfrak{p} \in V(I_\alpha),\ \forall\alpha \iff \mathfrak{p} \in \bigcap_\alpha I_\alpha.\end{aligned}$$

Exercise 5

First assume there exists $0, 1 \neq e \in A$ with $e^2 = e$. Then we claim that $\operatorname{Spec} A = V(e) \sqcup V(1-e)$: By exercise 4, $V(e) \cup V(1-e) = V(e(1-e)) = V(0) = \operatorname{Spec} A$, and $V(e) \cap V(1-e) = V((e)+(1-e)) = V(1) = \emptyset$. Also $V(e) \neq \emptyset$, since e is a zero-divisor, hence not a unit. Since also $(1-e)^2 = 1-e$, the same argument applies for $V(1-e)$ and the above is a non-trivial decomposition of $\operatorname{Spec} A$ into two clopen subsets.

For the converse, first note that $\operatorname{Spec} A$ is quasi-compact. Indeed, it suffices to consider coverings of the form $\operatorname{Spec} A = \bigcup_i D(f_i)$. This is equivalent to $(f_i)_i = 1$, but this means 1 is generated by finitely many of the f_i.

Now suppose $\operatorname{Spec} A = V(I) \sqcup V(J)$ is a non-trivial decomposition. By the same arguments as above, we have $\sqrt{0} \supseteq IJ$ and $1 = \sqrt{I+J} = I + J$. Since $V(I), V(J)$ are closed subsets of $\operatorname{Spec} A$, they are quasi-compact as well. So we can write $V(J) = \operatorname{Spec} A \setminus V(I) = \bigcup_i D(f_i)$ for finitely many i. Then $V(I) = V(f_i)_i$ and wlog we may assume that I, J are finitely generated.

Then IJ is finitely generated as well, and $IJ \subseteq \sqrt{0}$ implies $(IJ)^N = 0$ for some $N \gg 0$ (take the sum of exponents for a generating set). Raising $I + J = 1$ to the $2N-1$-th power, we see $I^N + J^N = 1$, so write $1 = e + (1-e)$ with $e \in I^N$ and $1 - e \in J^N$. Then $e(1-e) \in (IJ)^N = 0$, i.e. $e^2 = e$.

Exercise 1

Let $\operatorname{Spec} A = Z_1 \cup Z_2$ with Z_1, Z_2 closed. Suppose wlog that $(0) \in Z_1$. Then $Z_1 \supseteq \overline{\{(0)\}} = \operatorname{Spec} A$.

Exercise 2

(1) Last week.

(2) If $A = A_1 \times A_2$, then $e = (1, 0)$ is a nontrivial idempotent, so the claim follows from (1). Conversely, by (1) we have a nontrivial idempotent $e^2 = e \in A$. Let $\pi : A \to A/eA \times A/(1-e)A$ be the natural map. If $x \in \ker \pi$, then $x \in eA \cap (1-e)A = (e(1-e))A = 0$. Let $(a+eA, b+(1-e)A) \in A/eA \times A/(1-e)A$ be given. Then $\pi(a(1-e) + be) = (a(1-e) + eA, be + (1-e)A) = (a + eA, b + (1-e)A)$. So π is an isomorphism.

(3) Let $(A, \mathfrak{m})$ be local and assume A contains a non-trivial idempotent e. Then e is in particular a zero-divisor, hence not a unit, and the same applies to $1 - e$. Hence $e, 1 - e \in \mathfrak{m}$, but then also $1 = e + (1 - e) \in \mathfrak{m}$, contradiction.

Exercise 3

(1) The morphism $\pi : A \to S^{-1}A$ has the following universal property: For every morphism $f : A \to B$ such that $f(S) \subseteq B^\times$, there exists a unique morphism $\widetilde{f} : S^{-1}A \to B$ with $f = \widetilde{f} \circ \pi$.

(2) For the universal property to work, we only have to check that every element of $\{1, f, \dots, f^n, \dots\}$ is invertible in A_h. Clearly it suffices to do this for f. Since $D(h) \subseteq D(f) \iff V(f) \subseteq V(h)$, by Lemma 2.2 $h \in \sqrt{(f)}$, i.e. $h^m = af$ for some $a \in A$ and $m > 0$. Since h is invertible in A_h, so is f, and the result follows.

(3) These are exactly the restriction maps $\mathcal{O}(D(f)) \to \mathcal{O}(D(h))$.

Exercise 4

(1) We show more generally:

Lemma 1. *Let $\mathcal{C}$ be a category with direct limits, $(I, \leq)$ a directed set indexing a directed system $(A_i)_{i\in I}$, $(\varphi_{ij} : A_i \to A_j)_{i<j}$ in $\mathcal{C}$. Let $\iota_i : A_i \to \varinjlim_{i\in I} A_i =: A$ be its direct limit. Let $J \subseteq I$ be a* cofinal *subset, that is, for every $i \in I$ there exists $j \in J$ with $i \leq j$. Then J is directed and $(A_j)_{j\in J}, (\varphi_{ij})_{i<j;i,j\in J}$ is a directed system; write $\iota'_j : A_j \to \varinjlim_{j\in J} A_j =: A'$ for the direct limit of this sub-directed system. Then there exists a unique isomorphism $\varphi : A' \to A$ such that $\iota_j = \varphi \circ \iota'_j$ for every $j \in J$.*

Proof. $(\iota_j : A_j \to A)_{j\in J}$ satisfy $\iota_{j'} = \iota_j \circ \varphi_{jj'}$ by definition, so they induce a unique morphism $\varphi : A' \to A$ such that $\iota_j = \varphi \circ \iota'_j$. It remains to construct an inverse morphism. For $i \in I$ choose $j \in J$ with $i \leq j$, and set $\psi_i := \iota'_j \circ \varphi_{ij} : A_i \to A'$. Note that ψ_i does not depend on the choice of j: Let $j' \in J$ be another element with $i \leq j'$, since J is directed we may assume $j \leq j'$. (Otherwise pick $j'' \in J$ with $j \leq j''$ and $j' \leq j''$ and apply the result to both of these.) Then $\iota'_{j'} \circ \varphi_{ij'} = \iota'_{j'} \circ \varphi_{jj'} \circ \varphi_{ij} = \iota'_j \circ \varphi_{ij}$. Now let $i \leq k \in I$, and choose $J \ni j \geq k$. Then $\psi_k \circ \varphi_{ik} = \iota'_j \circ \varphi_{kj} \circ \varphi_{ik} = \iota'_j \circ \varphi_{ij} = \psi_i$, so the $(\psi_i)_{i\in I}$ induce a morphism $\psi : A \to A'$ with $\psi_i = \psi \circ \iota_i$.

To show $\psi \circ \varphi = \mathrm{id}_{A'}$, by uniqueness it suffices to show $\psi \circ \varphi \circ \iota'_j = \iota'_j$ for every $j \in J$. Indeed, $\psi \circ \varphi \circ \iota'_j = \psi \circ \iota_j = \psi_j = \iota'_j \circ \varphi_{jj} = \iota'_j$. Similarly, $\varphi \circ \psi \circ \iota_i = \varphi \circ \psi_i = \varphi \circ \iota'_j \circ \varphi_{ij} = \iota_j \circ \varphi_{ij} = \iota_i$ for every $i \in I$, where $i \leq j \in J$, implies $\varphi \circ \psi = \mathrm{id}_A$. □

Back to the exercise: The directed system in question is $(\{U \subseteq X \text{open} \mid x \in U\}, \supseteq)$ so by definition of a base of a topological space it is clear that $(\{U \in \mathcal{B} \mid x \in U\}, \supseteq)$ is a sub-directed set.

(2) From the proposition and (1), we immediately have

$$A_\rho \cong \mathcal{O}_{\operatorname{Spec} A, \rho} = \varinjlim_{U \ni \rho} \mathcal{O}_{\operatorname{Spec} A}(U) \cong \varinjlim_{D(f) \ni \rho} \mathcal{O}_{\operatorname{Spec} A}(D(f)) \cong \varinjlim_{D(f) \ni \rho} A_f.$$

By commutative algebra alone, for $f \notin \rho$, f is invertible in A_ρ by definition, so the universal property yields a unique morphism φ_f s.t. $\pi = \varphi_f \circ \pi_f$, where $\pi : A \to \mathbb{A}_\rho$, $\pi_f : A \to A_f$ are the canonical maps. Write further $\varphi_{ij} : A_f \to A_h$ for the maps constructed in exercise 3(2). The $(\varphi_{fh})_{D(f) \supseteq D(h)}$ is clearly a directed system, and $\varphi_h \circ \varphi_{fh}$ satisfies the universal property of φ_f, hence they are equal. Thus they induce a morphism $\varphi : \varinjlim_{D(f) \ni \rho} A_f \to A_\rho$, given by $[\frac{a}{f^n} \in A_f] \mapsto \frac{a}{f^n} \in A_\rho$. φ is clearly surjective, let $\alpha = [\frac{a}{f^n} \in A_f] \in \ker \varphi$. Then $as = 0$ for some $s \notin \rho$, hence $\frac{a}{f^n} = \frac{as}{f^n s} = 0$ in $A_{f^n s}$, i.e. $\alpha = 0$.

Exercise 1

(1) $f^{-1}(D(h)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in D(h)\} = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi(h) \notin \mathfrak{p}\} = D(\varphi(h))$.

(2) Let $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$. Using (1), we have a commutative diagram

$$\begin{tikzcd}
B \arrow[r, "\cong"] \arrow[d] & \mathcal{O}_X(X) \arrow[d] & \mathcal{O}_Y(Y) \arrow[l, "f_Y^\sharp"'] \arrow[d] & A \arrow[l, "\cong"] \arrow[lll, bend right, "\varphi"'] \arrow[d] \\
B_{\varphi(h)} \arrow[r, "\cong"] & \mathcal{O}_X(D(\varphi(h))) & \mathcal{O}_Y(D(h)) \arrow[l, "f^\sharp_{D(h)}"'] & A_h \arrow[l, "\cong"] \arrow[lll, dashed, bend left, "\varphi_h"]
\end{tikzcd}$$

where the vertical arrows are localizations/restrictions and the isomorphisms on the left and right are proposition 2.11. The diagram of solid arrows commutes: The central square by the definition of a morphism of sheaves, the outer squares by the proof of proposition 2.11, the upper part by proposition 2.18. Hence the composition along the bottom row satisfies the universal property of φ_h, so they agree. In other words, the map $f_D^\sharp(h)$ is, up to identifications of sections with A_h resp. $B_{\varphi(h)}$, exactly the localization morphism φ_h.

Exercise 2

Follows directly from proposition 1.11 and exercise 3.4(1).

Exercise 3

(1) The localization map induces an inclusion- and primality-preserving bijection between the ideals of $S^{-1}A$ and the ideals of A not intersecting S.

(2) $\operatorname{im} f = \{\mathfrak{p} \in \operatorname{Spec} A \mid \varphi(\mathfrak{p}) \in \operatorname{Spec} A_g\} \overset{(1)}{=} \{\mathfrak{p} \in \operatorname{Spec} A \mid g \notin \mathfrak{p}\} = D(g)$. In the same way one shows $f(D(h)) = D(h) \cap D(g)$, so f is an open continuous map, i.e. a homeomorphism onto its image.

(3) By (2) we have a map $(f, f^\sharp) : \operatorname{Spec} A_g \to (D(g), \mathcal{O}_{\operatorname{Spec} A}|_{D(g)})$, which is a homeomorphism on topological spaces. It remains to check that $f^\sharp$ is an isomorphism, for which by exercise 2 it suffices to consider principal opens $D(h)$, $D(h) \subseteq D(g)$. Then by exercise 1, $f^\sharp_{D(h)}$ is canonically identified with the localization map $A_h \to (A_g)_h$, which is an isomorphism since they satisfy the same universal property.

Exercise 4

Let $X = \bigcup_i U_i$ be a cover by affine opens, and $\varphi : A \to \mathcal{O}_X(X)$ be a ring morphism. Let $\varphi_i : A \to \mathcal{O}_X(X) \to \mathcal{O}_X(U_i) = \mathcal{O}_{U_i}(U_i)$ be the composition of φ and the restriction. Then by proposition 2.18, there exists a ***unique*** morphism of schemes $f_i : U_i \to Y$ corresponding to φ_i. Let $V \subseteq U_i \cap U_j$. Since $\rho_{U_i V} \circ \varphi_i = \rho_{U_j V} \circ \varphi_i$, also $f_i|_V = f_j|_V$. As such affine opens V cover $U_i \cap U_j$, we have $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap u_j}$ and we can apply proposition 2.25 to obtain a morphism $f : X \to Y$ with $f|_{U_i} = f_i$.

In particular, $\rho_{XU_i} \circ f_Y^\sharp = (f_i)_{U_i}^\sharp = \varphi_i = \rho_{XU_i} \circ \varphi$ for all i, i.e. $f_Y^\sharp = \varphi$ by the sheaf condition. Conversely, starting with $f : X \to Y$, we have to show that the above construction applied to $f_Y^\sharp$ yields back f. By proposition 2.18, $f|_{U_i} = f_i$, since they poth correspond to φ_i. Hence we are done by the uniqueness in proposition 2.25.